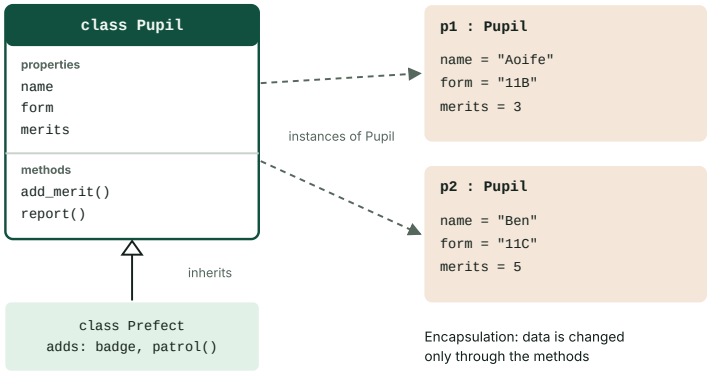
Programming paradigms: procedural and object-oriented

D01 ANSWERS

I can: Describe procedural programming as imperative, sequential step-by-step instructions · Describe object-oriented programming using objects, classes, properties and methods · Compare the two paradigms in an extended answer



Use this diagram to help you: A Pupil class, two objects created from it, and a Prefect class that inherits from it

A Key terms

Write the letter of the matching definition · 6 marks

A **Procedural programming**

E **Object-oriented programming**

B **Class**

F **Object**

D **Encapsulation**

C **Inheritance**

A Step-by-step instructions executed in sequence, organised into procedures and functions.

B A blueprint defining the properties and methods of objects.

C A class taking on the properties and methods of another class.

D Keeping an object's data inside it, accessed only through its methods.

E Programs built from self-contained objects that combine data (properties) and methods.

F An instance created from a class with its own data.

B Fill the gaps

Use the word bank · 7 marks

methods | inheritance | procedural | sequence | class | object | properties

In **procedural** programming the computer follows a **sequence** of instructions. In object-oriented programming a **class** is a template and an **object** is an instance of it. Data held by an object are its **properties** and its actions are its **methods**. A class that takes on the features of another class uses **inheritance**.

C True or false?

Tick one box · 4 marks

	True	False

Procedural languages are described as imperative. *(They issue commands.)*	☑	☐
In OOP, an object is a blueprint from which classes are created. *(A class is the blueprint; an object is an instance.)*	☐	☑
Encapsulation means an object's data can only be changed through its methods.	☑	☐
Object classes cannot be reused in other programs. *(Reuse is a key benefit of OOP.)*	☐	☑

D Exam-style questions

17 marks

1 Describe procedural programming. [3]

> Any three from: the programmer writes step-by-step instructions telling the computer exactly what to do (1) / it is imperative – each statement is a command (1) / instructions are executed sequentially, one after another, so the order is the program flow (1) / code is grouped into procedures and functions that can be called when needed (1) / the program is designed top-down – the main task is broken into smaller sub-tasks (1) / data is held in variables, kept separate from the code, and passed into procedures (1)

2 State the difference between a class and an object. [2]

> A class is a blueprint/template that defines the properties and methods for a type of object (1) / an object is a single instance created from a class, with its own data values (1). Accept e.g. class Pupil; objects p1 and p2.

3 In OOP, what is meant by a property and a method? [2]

> A property (attribute) is an item of data that an object holds, e.g. a Pupil's Name or Form (1) / a method is an action/subroutine an object can carry out, e.g. AddMerit() (1)

4 Explain one benefit of inheritance. [2]

> Inheritance lets a new class take on the properties and methods of an existing class, e.g. Prefect inherits from Pupil (1) / so code does not have to be rewritten – it is reused and only extended, making programs quicker to build and more reliable (1)

5 Describe encapsulation. [2]

> An object's data is kept (hidden) inside the object (1) / and can only be accessed or changed through the object's own methods, which protects the data from accidental change (1). Accept: data and the methods that act on it are bundled together in the object; private fields.

6 Describe procedural programming and object-oriented programming. (QWC) [6]

> Indicative content – Procedural: a sequence of step-by-step instructions telling the computer what to do; imperative (commands); executed sequentially in order; code grouped into procedures/functions that are called when needed; top-down design breaking the task into sub-tasks; data held in variables, separate from code, passed to procedures; suits smaller, linear tasks. Object-oriented: the program is built from self-contained objects holding both data and the code that works on it; a class is a blueprint and an object is an instance of a class; properties/attributes are the data, methods are the actions; objects communicate by calling each other's methods; encapsulation keeps data inside an object, changed only through its methods; inheritance lets a new class reuse and extend an existing class; classes can be reused in other programs; design starts with the objects; suits large, complex systems, GUIs and games. Levels (QWC): Level 3 (5–6 marks): thorough description of BOTH paradigms in similar depth (procedural beyond just 'step-by-step'); uses technical terms accurately; well-organised, fluent answer with few errors in spelling, punctuation and grammar. Level 2 (3–4 marks): description of both paradigms but one is thin, or one paradigm described well; some technical terms; reasonably organised with some errors. Level 1 (1–2 marks): a few basic points, e.g. 'step-by-step' or 'uses objects'; little or no technical vocabulary; poorly organised with frequent errors. 0 marks: nothing creditworthy.

E Quick quiz

Circle one answer · 5 marks

1 Which describes procedural programming? [1]

A Programs built from interacting objects

B Step-by-step instructions executed in sequence ✓

C Programs that only use a GUI

D Code that translates itself

2 A class is... [1]

A an instance of an object

B a blueprint for creating objects ✓

C a type of loop

D a syntax error

3 The actions an object can perform are its... [1]

A properties

B methods ✓

C parameters

D indexes

4 Which OOP feature lets a new class reuse an existing class's code? [1]

A Encapsulation

B Inheritance ✓

C Iteration

D Concatenation

5 Hiding an object's data so it can only be changed through its methods is... [1]

A encapsulation ✓

B decomposition

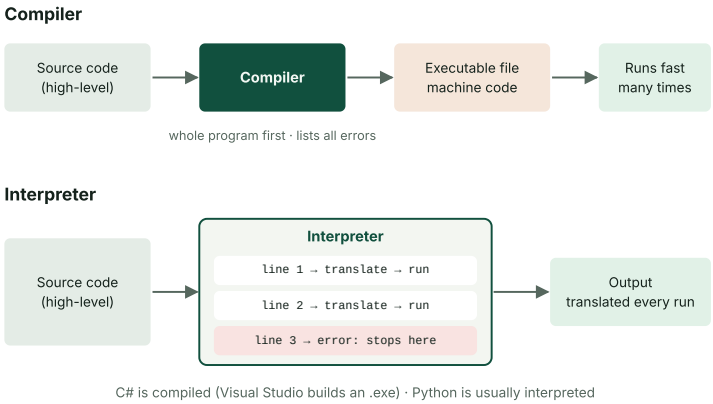
C abstraction

D compilation

Development environments, editing features and translators

D02 ANSWERS

I can: Describe the features of an IDE and the editing features of a code editor · Explain why high-level code must be translated into machine code · Compare compilers and interpreters



Use this diagram to help you: A compiler translates the whole program into an executable; an interpreter translates and runs one line at a time

A Key terms

Write the letter of the matching definition · 6 marks

D **IDE**

B **IntelliSense / auto-complete**

E **Translator**

F **Machine code**

A **Compiler**

C **Interpreter**

A Translates the whole program into machine code before running, producing a separate executable.

B Editor feature that suggests or completes code as it is typed.

C Translates and executes source code one line at a time.

D Integrated development environment: editor, translator, debugger and tools in one application.

E Software that converts source code into machine code: a compiler or interpreter.

F Binary instructions the processor executes directly.

B Fill the gaps

Use the word bank · 6 marks

interpreter | translator | IntelliSense | source | machine | compiler

Programs are written as **source** code in a high-level language. The processor can only run **machine** code, so a **translator** is needed. A **compiler** translates the whole program at once and creates an executable file. An **interpreter** translates and runs one line at a time. Editor features such as **IntelliSense** suggest code as the programmer types.

C True or false?

Tick one box · 4 marks

	True	False

An interpreter produces an executable file that can be run without it. *(A compiler does; an interpreter translates every run.)*	☐	☑
Compiled programs generally run faster than interpreted ones.	☑	☐
Line numbering helps a programmer find the line given in an error message.	☑	☐
A debugger lets the programmer step through code and watch variables.	☑	☐

D Exam-style questions

12 marks

1 Explain why source code written in a high-level language must be translated. [2]

> The CPU can only execute machine code (binary instructions) (1) / so high-level source code, which is close to English, must be converted into machine code by a translator (compiler or interpreter) before it can run (1)

2 Describe two editing features of an IDE and how each helps a programmer. [4]

> Any two features, each named (1) + how it helps (1), e.g.: IntelliSense/auto-complete (1) – suggests keywords, variable and function names and shows parameters as you type, so fewer typing and syntax errors and faster coding (1) / syntax highlighting/colour coding (1) – keywords, strings and comments in different colours so code is easier to read and mistakes stand out (1) / syntax error highlighting (wavy underline) (1) – errors are spotted immediately as you type (1) / line numbering (1) – find a line quickly when an error message gives a line number (1) / code outlining (1) – collapse/expand blocks so long programs are easier to navigate (1) / auto-indent (1) – keeps indentation consistent so structure is clear (1)

3 Describe how a compiler translates a program. [2]

> The compiler translates the whole source program into machine code in one go before it runs (1) / producing a separate executable file; it reports all errors as a list after compiling and will not produce the executable until syntax errors are fixed (1)

4 Give one advantage of using an interpreter while developing a program. [2]

> Any one advantage with expansion: it translates and runs one line at a time and stops as soon as it reaches an error (1) / showing exactly where the error is, so it is easier and quicker to debug while developing (1). Accept: no need to wait for the whole program to compile (1) / so changes can be tested immediately (1).

5 Give one reason why a software company would distribute a compiled program. [2]

> Any one reason with expansion: the source code is not distributed, only the executable (1) / so others cannot see, copy or alter the code – protects the company's intellectual property (1). OR: the compiled program runs quickly (1) / and users do not need a translator installed to run it (1).

E Quick quiz

Circle one answer · 5 marks

1 Which translator converts a whole program into machine code before it runs? [1]

A Interpreter — **B Compiler ✓**
C Debugger — D Editor

2 Which IDE feature suggests code and shows parameters while you type? [1]

A Line numbering — **B IntelliSense/auto-complete ✓**
C Code outlining — D Breakpoint

3 An interpreter... [1]

A translates one line at a time ✓ — B creates an .exe file
C only works on binary — D removes logic errors

4 The CPU can only execute... [1]

A pseudocode — **B machine code ✓**
C Python — D flowcharts

5 Collapsing and expanding blocks of code is called... [1]

A code outlining ✓

B syntax highlighting

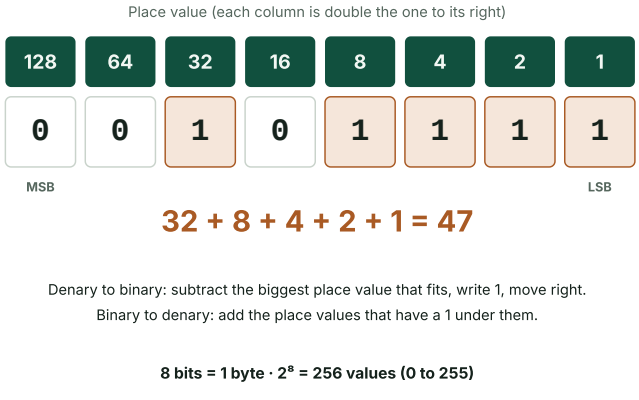
C compiling

D decomposition

Binary, units of data and denary–binary conversion

D03 ANSWERS

I can: Explain why computers store data in binary · Convert denary numbers to 8-bit binary and binary to denary · Use units of data and calculate storage capacities



Use this diagram to help you: Place values of an 8-bit byte showing 47 as 00101111

A Key terms

Write the letter of the matching definition · 6 marks

C **Denary**	**A** American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.
B **Binary**	**B** Base-2 number system using 0 and 1.
E **Hexadecimal**	**C** Base-10 number system.
F **Carry**	**D** When a result is too large for the available bits, so the extra bit is lost and the result is wrong.
D **Overflow**	**E** Base-16 number system (0–9, A–F); a shorthand for binary.
A **ASCII**	**F** The digit passed to the next column when a binary column total exceeds 1.

B Fill the gaps

Use the word bank · 6 marks

nibble | bit | byte | megabyte | 255 | 128

A single binary digit is called a **bit**. Four bits make a **nibble** and eight bits make a **byte**. In an 8-bit number the leftmost column is worth **128**. The largest number that can be stored in one byte is **255**. 1024 kilobytes make one **megabyte**.

C True or false?

Tick one box · 4 marks

	True	False
00001111 in binary is 15 in denary. *(8+4+2+1)*	☑	☐
A byte can store 256 different values. *(2^8 = 256)*	☑	☐
A nibble is 8 bits. *(A nibble is 4 bits.)*	☐	☑

1 terabyte = 1024 megabytes. *(1 TB = 1024 GB.)* ☐ ☑

D Exam-style questions

12 marks

1 Explain what is meant by a bit. [2]

> A bit is a binary digit (1) / the smallest unit of data, with a value of either 0 or 1 (off/on) (1)

2 Convert the denary number 54 into an 8-bit binary number. Show your working. [2]

> Working, e.g. 32 + 16 + 4 + 2 = 54 under place values 128 64 32 16 8 4 2 1 (or repeated division by 2 with remainders) (1) / 00110110 (1)

3 Convert the binary number 10101010 into denary. Show your working. [2]

> Working: 128 + 32 + 8 + 2 (1) / = 170 (1)

4 Convert 200 into binary. [2]

> Working, e.g. 128 + 64 + 8 = 200 (1) / 11001000 (1)

5 How many different values can be stored in 4 bits? [1]

> 16 (2^4 = 16) (1)

6 A disk holds 2 TB. It already stores 1990 GB. Will a 50 GB game fit? Show your working. [3]

> 2 TB = 2 × 1024 = 2048 GB (1) / free space = 2048 − 1990 = 58 GB (1) / 58 GB ≥ 50 GB, so yes, the game will fit (8 GB left over) (1). Accept answers using 1 TB = 1000 GB only if the conclusion follows from correct working (2000 − 1990 = 10 GB, so it would not fit) – course convention is 1024.

E Quick quiz

Circle one answer · 5 marks

1 What is 00010011 in denary? [1]

A 17 **B 19 ✓**
C 21 D 35

2 What is 37 in 8-bit binary? [1]

A 00100101 ✓ B 00100011
C 00110101 D 01000101

3 How many bits are in a nibble? [1]

A 2 **B 4 ✓**
C 8 D 16

4 Largest denary number in 8 bits? [1]

A 128 **B 255 ✓**
C 256 D 512

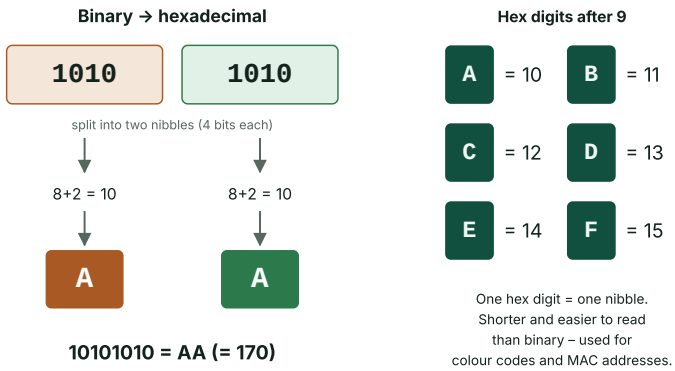
5 How many GB are in 3 TB? [1]

A 3000 **B 3072 ✓**
C 1024 D 30 720

Hexadecimal

D04 ANSWERS

I can: Explain what hexadecimal is and why it is used · Convert between binary and hexadecimal using nibbles · Convert between denary and hexadecimal



Use this diagram to help you: Converting the byte 10101010 to hexadecimal AA by splitting it into nibbles

A Key terms

Write the letter of the matching definition · 6 marks

C Hexadecimal

D Denary

A Binary

F Carry

B Overflow

E ASCII

A Base-2 number system using 0 and 1.

B When a result is too large for the available bits, so the extra bit is lost and the result is wrong.

C Base-16 number system (0–9, A–F); a shorthand for binary.

D Base-10 number system.

E American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.

F The digit passed to the next column when a binary column total exceeds 1.

B Fill the gaps

Use the word bank · 5 marks

nibble | 16 | A | F | two

Hexadecimal is base **16**. The letter **F** represents 15. Each hex digit represents one **nibble**, so a byte is written as **two** hex digits. The binary number 1010 is the hex digit **A**.

C True or false?

Tick one box · 4 marks

	True	False
Hex digit B represents 11.	☑	☐
Computers store data in hexadecimal instead of binary. *(Hex is a human-friendly shorthand; storage is binary.)*	☐	☑
FF in hexadecimal is 255 in denary.	☑	☐

One hex digit represents 8 bits. *(One hex digit = 4 bits.)* ☐ ☑

D Exam-style questions

12 marks

1 Convert the binary number 10101010 into hexadecimal. Show your working. [2]

> Split into nibbles: 1010 | 1010 = 10 | 10 (1) / AA (1)

2 Convert the denary number 66 into hexadecimal. Show your working. [3]

> Either method: 66 ÷ 16 = 4 remainder 2 (1) / the quotient 4 is the first digit and the remainder 2 the second (1) / 42 (1). Accept via binary: 66 = 01000010 (1) → 0100 | 0010 (1) → 42 (1).

3 Convert the hexadecimal number 3B into denary. [2]

> 3 × 16 + 11 (B = 11) = 48 + 11 (1) / = 59 (1)

4 Convert the hex number C4 into binary. [2]

> C = 1100 and 4 = 0100 (1) / 11000100 (1)

5 Give one advantage of using hexadecimal instead of binary. [1]

> Any one from: it is much shorter, so easier for humans to read and remember (1) / fewer mistakes when reading or copying long binary numbers (1) / easy to convert to and from binary (1)

6 Give two uses of hexadecimal in computing. [2]

> Any two from: colour codes in HTML/CSS, e.g. #1D6B53 (1) / MAC addresses (1) / memory addresses (1) / error codes (1)

E Quick quiz

Circle one answer · 5 marks

1 What is 1111 0000 in hex? [1]

A F0 ✓ B 0F
C FF D E0

2 Hex digit D represents... [1]

A 12 **B 13** ✓
C 14 D 15

3 2F in denary is... [1]

A 32 B 45
C 47 ✓ D 215

4 Why is hex used? [1]

A Computers can only store hex **B It is a shorter, easier-to-read form of binary** ✓
C It uses less memory D It encrypts data

5 80 in hexadecimal is... [1]

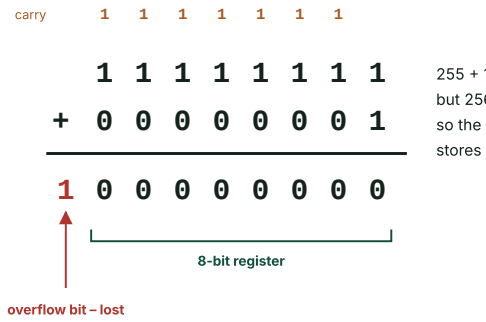
A 80 **B 50** ✓
C 48 D A0

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DATA

Binary addition and overflow

D05 ANSWERS

I can: Apply the four rules of binary addition with carries · Add two bytes and check the answer in denary · Explain overflow and its effect on a result



Rules: 0+0=0 · 0+1=1 · 1+1=0 carry 1 · 1+1+1=1 carry 1

Use this diagram to help you: Adding 11111111 and 00000001 in 8 bits causes overflow

A Key terms

Write the letter of the matching definition · 6 marks

A Carry	A The digit passed to the next column when a binary column total exceeds 1.
B Overflow	B When a result is too large for the available bits, so the extra bit is lost and the result is wrong.
C Denary	C Base-10 number system.
F Binary	D Base-16 number system (0–9, A–F); a shorthand for binary.
D Hexadecimal	E American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.
E ASCII	F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

8 | overflow | lost | carry | 255

In binary, 1 + 1 = 0 **carry** 1. When the result of an addition is too large for the available bits, **overflow** occurs. The largest value an **8** -bit register can hold is **255** . When overflow happens the extra bit is **lost** , so the stored result is incorrect.

C True or false?

Tick one box · 4 marks

	True	False
1 + 1 + 1 in binary gives 1 carry 1.	☑	☐
Overflow only happens with subtraction. *(It happens when any result is too large.)*	☐	☑
11111111 + 1 in 8 bits is stored as 00000000. *(The 9th bit is lost.)*	☑	☐

A carry out of the leftmost column means overflow has occurred. ☑ ☐

D Exam-style questions

12 marks

1 Add the binary numbers 00110101 and 00011010. Show the carries. [3]

> 00110101
> + 00011010
> = 01001111
> Carries shown into the 32 and 64 columns (1) / right-hand bits 1111 correct (1) / final answer 01001111 (1). Check: 53 + 26 = 79.

2 Add 10110110 and 01100101. Show your working. [3]

> 10110110
> + 01100101
> = 1 00011011
> Carries shown (carries out of the 4, 32, 64 and 128 columns) (1) / 8-bit result 00011011 (1) / the carry out of the 128 column is a 9th bit – overflow has occurred (1). Check: 182 + 101 = 283, which is > 255; the stored 8 bits are 27. Accept the 9-bit answer 100011011 with overflow identified.

3 Explain the meaning of overflow. [2]

> Overflow occurs when the result of a calculation is too large (1) / to be stored in the number of bits available, e.g. greater than 255 in an 8-bit register (1)

4 Describe the effect of overflow on the result of an 8-bit addition. [2]

> The extra (9th) bit carried out of the leftmost column cannot be stored and is lost (1) / so the stored result is incorrect, e.g. 255 + 1 is stored as 00000000 (0), which can cause wrong results or the program to fail (1)

5 Without adding in binary, decide whether 150 + 120 overflows an 8-bit register. Explain. [2]

> 150 + 120 = 270 (1) / which is greater than 255, the largest value an 8-bit register can hold, so it overflows (1)

E Quick quiz

Circle one answer · 5 marks

1 What is 1 + 1 in binary? [1]

A 2 — B 11
C 10 ✓ — D 01

2 00001111 + 00000001 = [1]

A 00010000 ✓ — B 00011111
C 00001110 — D 11110000

3 Overflow occurs when... [1]

A a number is negative — **B a result is too large for the available bits ✓**
C a carry is used — D two nibbles are added

4 Which 8-bit addition overflows? [1]

A 100 + 100 — B 200 + 50
C 128 + 127 — **D 200 + 60 ✓**

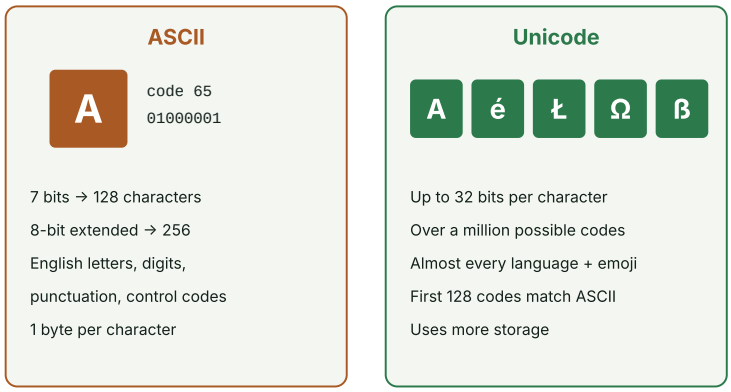
5 After overflow the stored result is... [1]

A correct — **B incorrect because a bit is lost ✓**
C always zero — D converted to hex

Character sets: ASCII and Unicode

D06 ANSWERS

I can: Explain what a character set is · Describe 7-bit ASCII, 8-bit (extended) ASCII and Unicode · Compare ASCII and Unicode



Use this diagram to help you: ASCII compared with Unicode

A Key terms

Write the letter of the matching definition · 6 marks

A ASCII	A American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.
D Unicode	B The digit passed to the next column when a binary column total exceeds 1.
C Denary	C Base-10 number system.
F Binary	D Character set covering almost all languages and emoji, using up to 32 bits per character; includes ASCII.
E Hexadecimal	E Base-16 number system (0–9, A–F); a shorthand for binary.
B Carry	F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

character set | 128 | 8 | more | Unicode

A **character set** gives each character a unique binary code. 7-bit ASCII can represent **128** characters. Extended ASCII uses **8** bits. **Unicode** can represent characters from almost every language, including emoji, but uses **more** bits per character.

C True or false?

Tick one box · 4 marks

	True	False
7-bit ASCII can represent 256 characters. *(128; 8-bit gives 256.)*	☐	☑
The first 128 Unicode characters match ASCII.	☑	☐
Unicode can represent emoji.	☑	☐

ASCII codes for capital letters are in alphabetical order. *(A=65, B=66...)* ☑ ☐

D Exam-style questions

11 marks

1 Expand the acronym ASCII. [1]

> American Standard Code for Information Interchange (1)

2 How many characters can be represented in 7-bit ASCII? Explain. [2]

> 128 characters (1) / because 7 bits give 2^7 = 128 different binary patterns (codes 0–127) (1)

3 The ASCII code for 'A' is 65. What is the code for 'E'? Give it in denary and binary. [2]

> E is 4 letters after A, so 65 + 4 = 69 (1) / 01000101 (1)

4 Describe two differences between ASCII and Unicode. [4]

> Two distinct differences, each stated (1) + expanded (1), e.g.: ASCII can represent only 128 (7-bit) or 256 (extended) characters (1) – Unicode can represent over a million characters, covering almost all world languages and emoji (1) / ASCII uses 7 or 8 bits per character (1) – Unicode uses more bits, up to 32 per character (UTF-8: 1–4 bytes), so text can take more storage (1). Do not credit the same difference written twice in reverse.

5 Explain why a company with customers worldwide would use Unicode. [2]

> Unicode gives every character in every language a unique code (including e.g. Chinese, Arabic, accented letters) (1) / so customers' names, addresses and messages in any language can be stored, displayed and exchanged correctly between countries and systems – ASCII mainly covers English (1)

E Quick quiz

Circle one answer · 5 marks

1 What does ASCII stand for? [1]

A American Standard Code for Information Interchange ✓ B Advanced System Code for Internet Input
C American System for Computer Information D Automatic Standard Character Index

2 How many characters in 7-bit ASCII? [1]

A 64 **B 128 ✓**
C 256 D 1024

3 Which can represent Chinese characters and emoji? [1]

A 7-bit ASCII B Extended ASCII
C Unicode ✓ D Hexadecimal

4 A drawback of Unicode compared with ASCII is... [1]

A fewer characters **B more bits per character, so more storage ✓**
C it only supports English D it cannot store digits

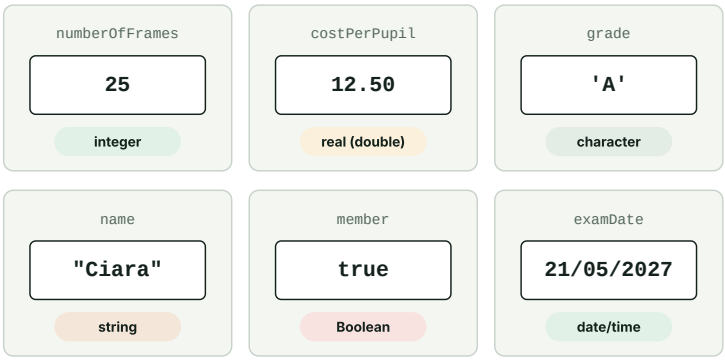
5 If 'a' = 97, what is 'c'? [1]

A 98 **B 99 ✓**
C 100 D 67

Data types

D07 ANSWERS

I can: Describe integer, real, character, string, Boolean and date/time data types · Choose and justify a data type for a variable · Use type conversion (casting) in C#



Console.ReadLine() returns a string – convert with Convert.ToInt32() or Convert.ToDouble() first.

Use this diagram to help you: Six variables, each holding a value of a different data type

A Key terms

Write the letter of the matching definition · 6 marks

D **Boolean**	**A** The digit passed to the next column when a binary column total exceeds 1.
B **Float / real**	**B** Data type for decimal numbers.
C **Denary**	**C** Base-10 number system.
F **Binary**	**D** Data type with only two values, True or False.
E **Hexadecimal**	**E** Base-16 number system (0–9, A–F); a shorthand for binary.
A **Carry**	**F** Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

convert · Boolean · integer · real · string

A whole number is stored as an **integer**. A number with a decimal part is stored as a **real**. A variable that can only be true or false is a **Boolean**. Text is stored as a **string**. In C#, Console.ReadLine() always returns a string, so Convert.ToInt32() is used to **convert** it.

C True or false?

Tick one box · 4 marks

	True	False
A phone number should be stored as an integer. *(String – leading zero, no maths.)*	☐	☑
In C#, Console.ReadLine() returns a string.	☑	☐
A Boolean can store three values. *(Only True or False.)*	☐	☑

The price £4.99 should be stored as a real. ☑ ☐

D Exam-style questions

10 marks

1 State the most suitable data type for: (a) number of nights booked (b) cost of a booking (c) whether equipment is hired (d) customer name. [4]

> (a) integer (1) / (b) real (accept float/double/currency) (1) / (c) Boolean (1) / (d) string (1)

2 Describe the Boolean data type. [1]

> A Boolean can hold only one of two values: True or False (1)

3 Explain why a race time of 12.84 seconds should be stored as a real. [1]

> The time has a decimal (fractional) part, which an integer cannot store, so a real is needed (1)

4 A C# program contains: string score = Console.ReadLine(); int total = score + 5; Explain the error and how to fix it. [2]

> Console.ReadLine() returns a string, so score is a string and cannot be added to the integer 5 – type mismatch, so the program will not compile (1) / fix: convert it first, e.g. int total = Convert.ToInt32(score) + 5; (accept int.Parse(score), or declare int score = Convert.ToInt32(Console.ReadLine());) (1)

5 Describe the difference between a character and a string. [2]

> A character holds a single letter, digit or symbol, e.g. 'A' (1) / a string holds a sequence of characters (text), e.g. "Ciara" (1)

E Quick quiz

Circle one answer · 5 marks

1 Best data type for the number of pupils on a trip? [1]

A Real — **B Integer ✓**
C Boolean — D String

2 Best data type for 'Has the pupil paid?' [1]

A Boolean ✓ — B Character
C Integer — D Date/time

3 Which C# method converts "12" into a whole number? [1]

A Convert.ToString() — B Convert.ToDouble()
C Convert.ToInt32() ✓ — D Console.ReadLine()

4 What is the value of 7 / 2 in C# when both are int? [1]

A 3.5 — **B 3 ✓**
C 4 — D Error

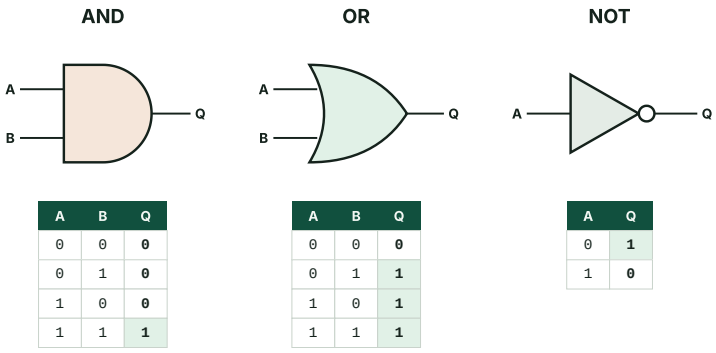
5 Best data type for a customer's surname? [1]

A Character — **B String ✓**
C Real — D Boolean

Boolean operators and truth tables

D08 ANSWERS

I can: Describe the AND, OR and NOT operators · Complete truth tables for single and combined expressions · Use Boolean operators in program conditions



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

A	Q
0	1
1	0

AND: 1 only if both inputs are 1 · OR: 1 if at least one is 1 · NOT: reverses the input

Use this diagram to help you: AND, OR and NOT logic gates with their truth tables

A Key terms

Write the letter of the matching definition · 6 marks

B Truth table

C AND / OR / NOT

D Denary

F Binary

E Hexadecimal

A Carry

A The digit passed to the next column when a binary column total exceeds 1.

B Table showing the output for every combination of inputs.

C AND is true only if both inputs are true; OR is true if at least one is; NOT reverses the input.

D Base-10 number system.

E Base-16 number system (0–9, A–F); a shorthand for binary.

F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

8 | OR | NOT | AND | truth table

The **AND** operator is true only when both inputs are true. The **OR** operator is false only when both inputs are false. **NOT** reverses its input. A **truth table** shows the output for every combination of inputs. With three inputs there are **8** rows.

C True or false?

Tick one box · 4 marks

	True	False
1 OR 0 = 1	☑	☐
1 AND 0 = 1 *(AND needs both true.)*	☐	☑
NOT 1 = 0	☑	☐

A truth table with two inputs has 4 rows. *($2^2 = 4$)* ☑ ☐

D Exam-style questions

12 marks

1 Complete the truth table for A AND B for all four input combinations. [2]

> A B | A AND B: 0 0 → 0 / 0 1 → 0 / 1 0 → 0 / 1 1 → 1. Award: first three rows all 0 (1) / 1 1 → 1 (1)

2 Complete the truth table for NOT (P AND Q). [2]

> P Q | P AND Q | NOT (P AND Q): 0 0 → 0 → 1 / 0 1 → 0 → 1 / 1 0 → 0 → 1 / 1 1 → 1 → 0. Award: 1, 1, 1 for the first three rows (1) / 0 for P = 1, Q = 1 (1)

3 How many rows does a truth table with three inputs need? Explain. [2]

> 8 rows (1) / because each input can be 0 or 1, so there are $2 \times 2 \times 2 = 2^3 = 8$ combinations (1)

4 Complete the output column for (P OR Q) AND R. [4]

> P Q R → output: 000 → 0, 001 → 0, 010 → 0, 011 → 1, 100 → 0, 101 → 1, 110 → 0, 111 → 1 (P OR Q column: 0 0 1 1 1 1 1 1). One mark per two correct rows: rows 1–2 (1) / rows 3–4 (1) / rows 5–6 (1) / rows 7–8 (1)

5 Write a condition that is True when a mark is between 50 and 100 inclusive. [2]

> mark >= 50 AND mark <= 100 (C#: mark >= 50 && mark <= 100). Award: both comparisons correct with >= and <= (1) / joined with AND (1). Do not accept OR.

E Quick quiz

Circle one answer · 5 marks

1 1 AND 1 = [1]

A 0 — **B 1 ✓**

C 2 — D 10

2 0 OR 0 = [1]

A 0 ✓ — B 1

C True — D Error

3 NOT (1 AND 0) = [1]

A 0 — **B 1 ✓**

C 10 — D None

4 Which condition is true only for ages 13 to 19? [1]

A age > 13 || age < 19 — **B age >= 13 && age <= 19 ✓**

C !(age == 13) — D age >= 13 || age <= 19

5 How many rows for 4 inputs? [1]

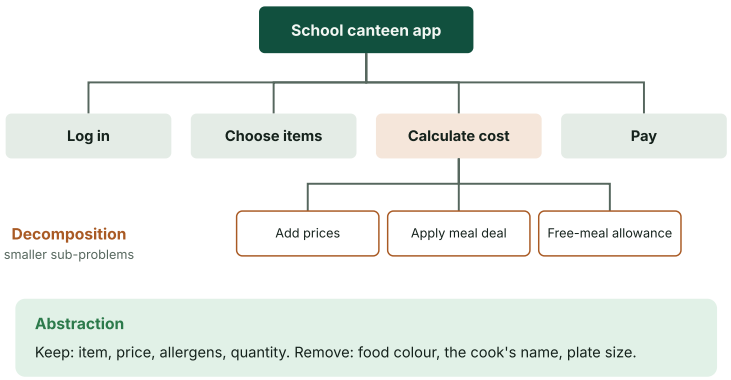
A 8 — B 12

C 16 ✓ — D 4

Computational thinking: abstraction and decomposition

D09 ANSWERS

I can: Define abstraction and decomposition · Apply both to a real problem · Explain the benefits of each for programmers



Use this diagram to help you: Decomposition of a school canteen app into sub-problems, with an abstraction note

A Key terms

Write the letter of the matching definition · 6 marks

E **Top-down design**

D **Abstraction**

A **Decomposition**

F **Algorithm**

B **Pseudocode**

C **Flowchart**

A Breaking a complex problem into smaller, more manageable sub-problems.

B Language-independent, English-like instructions describing an algorithm.

C Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision.

D Removing details of a problem that are not needed for the solution.

E Breaking the main task into smaller sub-tasks.

F A step-by-step plan for solving a problem.

B Fill the gaps

Use the word bank · 5 marks

Abstraction | algorithm | Decomposition | sub-problems | structure diagram

Abstraction means removing details that are not needed. **Decomposition** means breaking a problem into smaller **sub-problems**. A **structure diagram** shows top-down design. The final step-by-step solution is an **algorithm**.

C True or false?

Tick one box · 4 marks

	True	False
Abstraction means adding extra detail to a problem. *(It removes unnecessary detail.)*	☐	☑
Decomposition lets different programmers work on different parts.	☑	☐
A map is an example of abstraction.	☑	☐

Decomposition makes testing harder. *(Each part can be tested on its own.)* ☐ ☑

D Exam-style questions

14 marks

1 Define abstraction. [2]

> Removing or hiding details that are not needed to solve the problem (1) / so you focus only on what is important and the problem becomes simpler (1)

2 Define decomposition. [2]

> Breaking a large, complex problem down (1) / into smaller, more manageable sub-problems that can each be solved (and tested) separately (1)

3 A cinema wants a booking program. Give two details that could be removed through abstraction. [2]

> Any two relevant details, e.g.: the customer's favourite colour (1) / the colour of the seats/carpet (1) / the plot or actors of the film (1) / the customer's hair colour or clothing (1) / the cinema building's address history (1). Accept any detail not needed for a booking; do not accept film, time, seat, number of tickets or price.

4 Give two benefits of decomposition when developing a program. [2]

> Any two from: each sub-problem is easier to understand and solve (1) / work can be shared across a team, working at the same time (1) / parts can be reused (1) / each part can be tested separately, so errors are easier to find (1)

5 Explain abstraction and decomposition and the benefits of each. (QWC) [6]

> Indicative content – Abstraction: removing/hiding unnecessary detail to focus on what matters; a model keeps only what is needed; example in context (e.g. a ticket program needs age and journey type, not favourite colour; a route map shows stations, not buildings); benefits: simpler problem, less data to store and less code to write, quicker to develop. Decomposition: breaking a large problem into smaller sub-problems; often shown as a structure diagram (top-down design); example (e.g. canteen app: log in → view menu → choose items → calculate cost → pay); benefits: each part easier to understand and solve, work shared across a team, parts reused, each module tested separately so errors easier to find. Together they lead to an algorithm. Levels (QWC): Level 3 (5–6 marks): both terms clearly defined with examples and benefits of each; uses technical terms accurately; well-organised, fluent answer with few errors in spelling, punctuation and grammar. Level 2 (3–4 marks): both defined with some benefits, or one fully explained; some technical terms; reasonably organised with some errors. Level 1 (1–2 marks): basic definition of one or both terms; little or no technical vocabulary; poorly organised with frequent errors. 0 marks: nothing creditworthy.

E Quick quiz

Circle one answer · 5 marks

1 Removing unnecessary detail is... [1]

A decomposition
B abstraction ✓
C iteration
D validation

2 Breaking a problem into sub-problems is... [1]

A decomposition ✓
B abstraction
C compilation
D encryption

3 Which diagram shows decomposition? [1]

A Truth table
B Structure diagram ✓
C Trace table
D Pie chart

4 A benefit of decomposition is... [1]

A programs never have errors
B parts can be developed and tested separately ✓
C no algorithm is needed
D it removes all data

5 The London Underground map is an example of... [1]

A abstraction ✓

B overflow

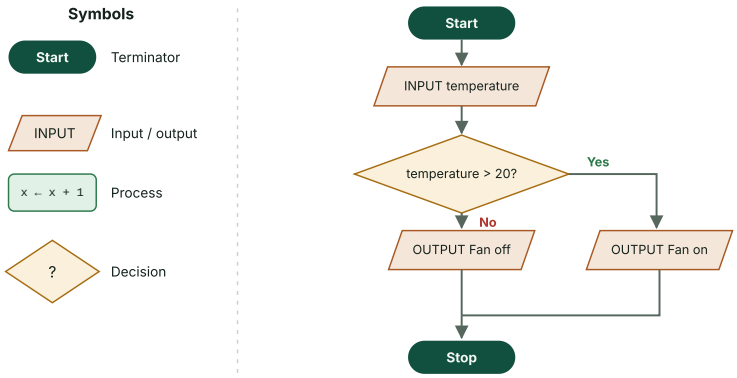
C iteration

D a protocol

Designing algorithms: flowcharts and pseudocode

D10 ANSWERS

I can: Define an algorithm and pseudocode · Use the standard flowchart symbols · Write and refine algorithms in pseudocode using sequence, selection and iteration



Use this diagram to help you: Flowchart symbols and a flowchart that turns a fan on above 20 degrees

A Key terms

Write the letter of the matching definition · 6 marks

C Algorithm

B Pseudocode

D Flowchart

F Top-down design

E Abstraction

A Decomposition

A Breaking a complex problem into smaller, more manageable sub-problems.

B Language-independent, English-like instructions describing an algorithm.

C A step-by-step plan for solving a problem.

D Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision.

E Removing details of a problem that are not needed for the solution.

F Breaking the main task into smaller sub-tasks.

B Fill the gaps

Use the word bank · 5 marks

algorithm | decision | refinement | parallelogram | Pseudocode

An **algorithm** is a step-by-step solution to a problem. In a flowchart a **decision** is shown by a diamond and an input or output by a **parallelogram**. **Pseudocode** is English-like and not tied to a language. Improving a design after testing it is called **refinement**.

C True or false?

Tick one box · 4 marks

	True	False
A diamond in a flowchart represents a process. *(It represents a decision.)*	☐	☑
Pseudocode must follow exact C# syntax. *(Pseudocode is not tied to any language.)*	☐	☑
Algorithms use sequence, selection and iteration.	☑	☐

A terminator shows the start or end of a flowchart. ☑ ☐

D Exam-style questions

17 marks

1 Define the term algorithm. [2]

> A precise, step-by-step set of instructions (1) / to solve a problem / carry out a task (1)

2 Describe what is meant by pseudocode. [2]

> An English-like (structured English) description of an algorithm using programming-style keywords (1) / that is language-independent and ignores exact syntax, so it can be converted into any language (1)

3 Name and describe the purpose of two flowchart symbols. [4]

> Two symbols, each named (1) + purpose (1), e.g.: terminator (rounded rectangle) (1) – start/stop of the algorithm (1) / input/output (parallelogram) (1) – input a value or output a result (1) / process (rectangle) (1) – a calculation or assignment (1) / decision (diamond) (1) – a Yes/No question with two exits (1) / flow line (arrow) (1) – shows the direction of flow (1) / subroutine (double-sided rectangle) (1) – calls a predefined procedure/function (1)

4 Write pseudocode that inputs a temperature and outputs 'Fan on' if it is above 20, otherwise 'Fan off'. [3]

```
INPUT temperature
IF temperature > 20 THEN
   OUTPUT "Fan on"
ELSE
   OUTPUT "Fan off"
END IF
```

> Marks: input the temperature (1) / correct condition temperature > 20 with 'Fan on' (1) / ELSE outputting 'Fan off' (1). Accept equivalent C#.

5 Write pseudocode to input five distances in miles, calculate the total and output the total in kilometres (1 mile = 1.609 km). [6]

```
total ← 0
FOR count ← 1 TO 5
   INPUT distance
   total ← total + distance
NEXT count
km ← total * 1.609
OUTPUT km
```

> Marks: total initialised to 0 before the loop (1) / loop that repeats 5 times (1) / input distance inside the loop (1) / total ← total + distance inside the loop (1) / conversion total * 1.609 after the loop (1) / output the total in km after the loop (1). Accept a constant for 1.609 and equivalent C#.

E Quick quiz

Circle one answer · 5 marks

1 Which flowchart symbol represents a decision? [1]

A Rectangle — **B Diamond ✓**

C Parallelogram — D Oval

2 Pseudocode is... [1]

A machine code — **B a language-independent description of an algorithm ✓**

C only used in C# — D a type of flowchart

3 In pseudocode, total ← total + 5 is... [1]

A selection — **B an assignment ✓**

C a comparison — D an input

4 Which symbol is used for INPUT and OUTPUT? [1]

A Parallelogram ✓ B Diamond

C Terminator D Arrow

5 Repeating a set of steps is called... [1]

A sequence B selection

C iteration ✓ D abstraction

Sorting algorithms: bubble sort and insertion sort

D11 ANSWERS

I can: Describe how a bubble sort and an insertion sort work · Show each pass of a sort on a list of values · Compare the efficiency of the two sorts



Use this diagram to help you: Bubble sort passes on 5, 3, 8, 1, 4

A Key terms

Write the letter of the matching definition · 6 marks

E **Bubble sort**	A Breaking the main task into smaller sub-tasks.
B **Insertion sort**	B Builds a sorted list by inserting each item into its correct position.
A **Top-down design**	C A step-by-step plan for solving a problem.
F **Abstraction**	D Breaking a complex problem into smaller, more manageable sub-problems.
D **Decomposition**	E Compares adjacent items and swaps them if out of order, repeating passes until no swaps occur.
C **Algorithm**	F Removing details of a problem that are not needed for the solution.

B Fill the gaps

Use the word bank · 5 marks

pass | bubble | insertion | adjacent | swaps

A **bubble** sort compares **adjacent** items and swaps them if they are in the wrong order. Each run through the list is called a **pass**. The sort stops when a pass makes no **swaps**. An **insertion** sort builds a sorted list by placing each item in its correct position.

C True or false?

Tick one box · 4 marks

	True	False
After the first pass of an ascending bubble sort the largest value is at the end.	☑	☐
Insertion sort needs the data to be sorted before it starts. *(That is binary search.)*	☐	☑
A bubble sort compares items that are next to each other.	☑	☐

Bubble sort is very efficient for large lists. *(It is slow for large lists.)*	☐	☑

D Exam-style questions

18 marks

1 Describe how a bubble sort orders a list into ascending order. [4]

> Any four from: compare the first two items (adjacent pair) (1) / if they are in the wrong order (first larger than second), swap them (1) / move along one position and compare the next pair, repeating to the end of the list – one pass (1) / after each pass the largest remaining item has 'bubbled' to its correct place at the end (1) / repeat passes until a whole pass makes no swaps (or n − 1 passes) (1)

2 Show the list after each pass of a bubble sort on 7, 2, 9, 4, 1. [4]

> Pass 1: 2, 7, 4, 1, 9 (1) / Pass 2: 2, 4, 1, 7, 9 (1) / Pass 3: 2, 1, 4, 7, 9 (1) / Pass 4: 1, 2, 4, 7, 9 (1). (A fifth pass makes no swaps, confirming the list is sorted – not required.)

3 Describe how an insertion sort works. [3]

> Any three from: the first item is treated as a sorted sub-list (1) / the next item is taken from the unsorted part (1) / and inserted into its correct position in the sorted part (1) / larger items are shifted one place right to make room (1) / repeat until every item has been inserted (1)

4 How does a bubble sort know the list is sorted? [1]

> A complete pass is made with no swaps (accept: a swapped flag stays false) (1)

5 Describe the bubble sort and insertion sort and compare their efficiency. (QWC) [6]

> Indicative content – Bubble sort: compares adjacent pairs and swaps if in the wrong order; one pass moves the largest remaining item to the end; repeat passes until a pass has no swaps (max n − 1 passes); swapped flag allows early stop. Insertion sort: first item is a sorted sub-list; each next item is taken and inserted into the correct position, shifting larger items right; like sorting a hand of cards. Efficiency: bubble sort is slow on large lists because of many comparisons and swaps; insertion sort is usually faster and very quick on nearly sorted data; both are simple to code and sort in place (no extra memory). Levels (QWC): Level 3 (5–6 marks): both sorts clearly described and a valid efficiency comparison made; uses technical terms accurately; well-organised, fluent answer with few errors in spelling, punctuation and grammar. Level 2 (3–4 marks): both sorts described with limited comparison, or one described well with comparison; some technical terms; reasonably organised with some errors. Level 1 (1–2 marks): basic points about one or both sorts; little or no technical vocabulary; poorly organised with frequent errors. 0 marks: nothing creditworthy.

E Quick quiz

Circle one answer · 5 marks

1 In a bubble sort, which items are compared? [1]

A First and last
B Adjacent items ✓
C Random pairs
D Middle item and target

2 After pass 1 of an ascending bubble sort on 4 2 5 1: [1]

A 2 4 1 5 ✓
B 1 2 4 5
C 2 1 4 5
D 4 2 1 5

3 When does a bubble sort stop? [1]

A After one pass
B When a pass makes no swaps ✓
C When the first item is smallest
D Never

4 Insertion sort works by... [1]

A halving the list
B inserting each item into its place in a sorted sub-list ✓
C swapping the first and last items
D searching for the middle item

5 Which is usually faster on nearly sorted data? [1]

A Bubble sort
B Insertion sort ✓

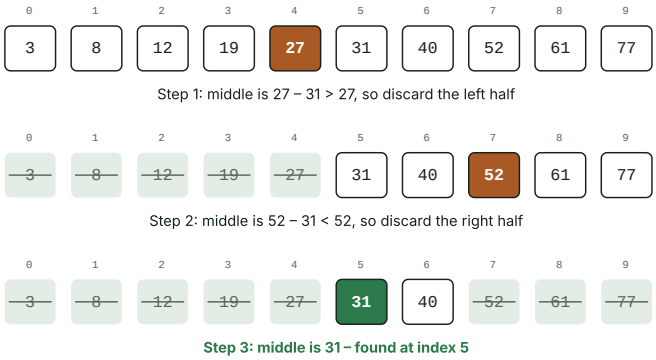
C Both are identical

D Linear search

Searching algorithms: linear and binary search

D12 ANSWERS

I can: Describe linear and binary search · Trace a binary search to find or not find an item · Compare the efficiency and requirements of each



Use this diagram to help you: Binary search for 31 in a sorted list, halving the search each step

A Key terms

Write the letter of the matching definition · 6 marks

E **Linear search**	A Breaking a complex problem into smaller, more manageable sub-problems.
B **Binary search**	B Repeatedly checks the middle of a sorted list and discards half; needs sorted data.
D **Top-down design**	C A step-by-step plan for solving a problem.
F **Abstraction**	D Breaking the main task into smaller sub-tasks.
A **Decomposition**	E Checks each item in turn until the target is found or the list ends.
C **Algorithm**	F Removing details of a problem that are not needed for the solution.

B Fill the gaps

Use the word bank · 5 marks

sorted | binary | linear | faster | middle

A **linear** search checks each item in turn. A **binary** search only works if the data is **sorted**. It looks at the **middle** item and discards half the list each time, so it is much **faster** for large lists.

C True or false?

Tick one box · 4 marks

	True	False
Linear search works on unsorted data.	☑	☐
Binary search checks every item one by one. *(It halves the list.)*	☐	☑
Binary search needs sorted data.	☑	☐

For a very short list, a linear search is perfectly suitable.	☑	☐

D Exam-style questions

11 marks

1 Describe how a linear search finds an item. [2]

> Start at the first item and check each item in turn (1) / until the target is found or the end of the list is reached (1)

2 Describe how a binary search finds an item. [3]

> Look at the middle item; if it is the target, it is found (1) / if the target is smaller discard the upper half, if larger discard the lower half (1) / repeat on the remaining half until found or no items are left (1)

3 State the condition that must be met before a binary search can be used. [1]

> The data must be sorted (in order) first (1)

4 Give two advantages of a binary search over a linear search. [2]

> Any two from: it is faster/more efficient for large lists (1) / because each comparison halves the number of items left (1) / so far fewer comparisons are needed, e.g. about 10 checks for 1000 items compared with up to 1000 for linear (1)

5 Trace a binary search for 40 in: 3 8 12 19 27 31 40 52 61 77. Show each middle value. [3]

> Middle = 27 (index 4); 40 > 27, so keep the upper half 31 40 52 61 77 (1) / middle = 52; 40 < 52, so keep 31 40; middle = 31; 40 > 31, keep 40 (1) / middle = 40 – found (1). Accept a consistent method that rounds the midpoint up (31 → 61 → 52 → 40 found).

E Quick quiz

Circle one answer · 5 marks

1 Which search requires sorted data? [1]

A Linear — **B Binary ✓**
C Both — D Neither

2 The first item a binary search checks is... [1]

A the first — B the last
C the middle ✓ — D a random item

3 Searching 1 000 items, which is likely fastest? [1]

A Linear — **B Binary ✓**
C Bubble sort — D Insertion sort

4 Linear search on [4, 9, 2, 7] for 7 needs how many comparisons? [1]

A 1 — B 2
C 3 — **D 4 ✓**

5 After checking the middle item and finding the target is larger, a binary search... [1]

A stops — **B discards the lower half ✓**
C discards the upper half — D restarts

Dry runs, trace tables, data requirements and interfaces

D13 ANSWERS

I can: Carry out a dry run using a trace table · Predict the output of an algorithm, including loops · Identify data requirements and design a simple user interface

```
a ← 1
WHILE a < 20
    a ← a * 2
END WHILE
OUTPUT a
```

Check the condition every time, including the last one: 32 < 20 is False, so the loop stops and 32 is output – not 16.

a	a < 20 ?	Output
1	True	
2	True	
4	True	
8	True	
16	True	
32	**False**	**32**

Use this diagram to help you: A dry run of a WHILE loop recorded in a trace table

A Key terms

Write the letter of the matching definition · 6 marks

A Dry run

E Trace table

C Desk checking

F Top-down design

D Abstraction

B Decomposition

A Working through code by hand, line by line, recording variable values.

B Breaking a complex problem into smaller, more manageable sub-problems.

C Reviewing code by hand to find errors before running it.

D Removing details of a problem that are not needed for the solution.

E Table recording the values of variables at each step of a dry run.

F Breaking the main task into smaller sub-tasks.

B Fill the gaps

Use the word bank · 5 marks

dry run | prompts | trace table | logic | data requirements

Working through an algorithm by hand is a **dry run** . The values of each variable are recorded in a **trace table** . Dry runs help find **logic** errors. Before coding, the designer identifies the **data requirements**, including each variable's data type. A good interface gives clear **prompts** .

C True or false?

Tick one box · 4 marks

	True	False
A dry run needs the program to be compiled first. *(It is done by hand.)*	☐	☑
Trace tables record variable values as the algorithm runs.	☑	☐

In a nested loop the inner loop runs fully for each pass of the outer loop.	☑	☐
Data requirements only list the program's outputs. *(Inputs, processing and outputs.)*	☐	☑

D Exam-style questions

10 marks

1 Explain how a dry run helps when testing a program. [2]

> Working through the algorithm/program by hand, line by line, without a computer (1) / to find logic errors and check the output is as expected (checking every path) (1)

2 What is a trace table? [1]

> A table that records the value of each variable (and the condition and any output) after each step of an algorithm (1)

3 Predict the output of: for (int i = 1; i < 4; i++) { Console.WriteLine(i * i); } [2]

> The loop runs for i = 1, 2, 3 (it stops when i = 4 because i < 4 is false) (1) / output 1, 4, 9, each on a new line, and nothing else (1). No credit for the second mark if 16 is included.

4 Complete a trace table for: total ← 0; FOR n ← 1 TO 4: total ← total + n; NEXT n; OUTPUT total. State the final output. [3]

> n: 1, 2, 3, 4 with total: 1, 3, 6, 10 (initial total 0). Award: total values 1, 3 correct (1) / total values 6, 10 correct (1) / final output 10 (1)

5 A hire program asks for number of nights, whether equipment is needed and the cost per night. Identify a suitable data type and validation rule for number of nights. [2]

> Integer (1) / range check, e.g. 1 to 14 nights (accept presence/type check, or > 0) (1)

E Quick quiz

Circle one answer · 5 marks

1 What is the output? x ← 3; WHILE x < 30: x ← x * 3; OUTPUT x [1]

A 9 — B 27
C 81 ✓ — D 30

2 A trace table records... [1]

A CPU speed — **B variable values at each step ✓**
C test plan results only — D file sizes

3 for (int i = 2; i < 5; i++) Console.Write(i + " "); outputs... [1]

A 2 3 4 5 — **B 2 3 4 ✓**
C 1 2 3 4 — D 3 4 5

4 A dry run is mainly used to find... [1]

A syntax errors — **B logic errors ✓**
C hardware faults — D viruses

5 Which belongs in data requirements? [1]

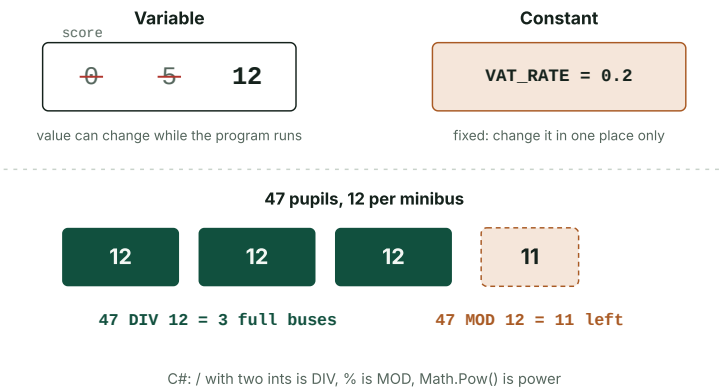
A Variable names and data types ✓ — B The colour of the PC
C The CPU clock speed — D The programmer's name

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Variables, constants, operators, input, output and assignment

D14 ANSWERS

I can: Explain the difference between a variable and a constant and the benefits of constants · Use input, output and assignment statements · Use arithmetic operators including DIV, MOD and powers



Use this diagram to help you: A variable whose value changes compared with a constant, and DIV and MOD

A Key terms

Write the letter of the matching definition · 6 marks

A Variable	A A named memory location whose value can change while the program runs.
E Constant	B Repeating a block of code (loops).
C Sequence	C Statements executed one after another in order.
F Selection	D A loop whose condition never becomes false, so it never ends.
B Iteration	E A named value that stays the same while the program runs.
D Infinite loop	F A condition decides which statements run (IF statements).

B Fill the gaps

Use the word bank · 5 marks

constant | assignment | DIV | variable | MOD

A **variable** is a named memory location whose value can change. A **constant** keeps the same value while the program runs. Storing a value in a variable is called **assignment**. The **MOD** operator gives the remainder of a division and **DIV** gives the whole-number part.

C True or false?

Tick one box · 4 marks

	True	False
17 MOD 5 = 2	☑	☐
A constant can be changed by the user while the program runs.	☐	☑

In C#, == is used to compare two values.	☑	☐
Variables should be given meaningful names.	☑	☐

D Exam-style questions

10 marks

1 Describe the difference between a variable and a constant. [2]

> A variable is a named memory location whose value can change while the program runs (1) / a constant is a named value that does not change while the program runs (1)

2 Give two advantages of using a constant in a program. [2]

> Any two from: improves readability – a meaningful name instead of a 'magic number' (1) / easier maintenance – the value only needs changing in one place (1) / prevents the value being changed by mistake while the program runs (1). Do not credit a definition of a constant.

3 What is the result of 23 MOD 5 and 23 DIV 5? [2]

> 23 MOD 5 = 3 (1) / 23 DIV 5 = 4 (1)

4 Write a pseudocode statement that assigns the result of price multiplied by quantity to a variable called total. [1]

> total ← price * quantity (1). Accept total = price * quantity.

5 Write pseudocode to input the length and width of a room and output its area. [3]

> INPUT length
> INPUT width
> area ← length * width
> OUTPUT area
> Marks: both inputs (1) / area ← length * width (1) / output area (1)

E Quick quiz

Circle one answer · 5 marks

1 Which value should be a constant? [1]

A A pupil's score
B The VAT rate ✓
C A running total
D The user's name

2 What is 14 MOD 4? [1]

A 3
B 2 ✓
C 3.5
D 1

3 What is 14 DIV 4? [1]

A 3 ✓
B 2
C 3.5
D 4

4 Which is an assignment? [1]

A OUTPUT total
B total ← total + 1 ✓
C IF total > 10
D INPUT total

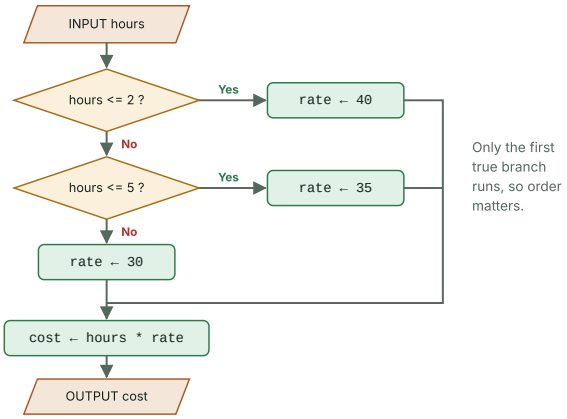
5 An advantage of a constant is... [1]

A it changes automatically
B it only needs updating in one place ✓
C it uses no memory
D it speeds up the CPU

Selection: IF, ELSE IF and nested IF

D15 ANSWERS

I can: Describe sequence and selection · Write IF, IF–ELSE and multi-branch selection in pseudocode and C# · Write tiered-pricing algorithms using single-condition IFs



Use this diagram to help you: Flowchart of an IF, ELSE IF, ELSE price calculation

A Key terms

Write the letter of the matching definition · 6 marks

A **Sequence**	**A** Statements executed one after another in order.
E **Selection**	**B** A loop whose condition never becomes false, so it never ends.
D **Variable**	**C** Repeating a block of code (loops).
F **Constant**	**D** A named memory location whose value can change while the program runs.
C **Iteration**	**E** A condition decides which statements run (IF statements).
B **Infinite loop**	**F** A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

Selection | condition | ELSE | sequence | nested

Selection allows a program to choose which statements to run. The block runs only when the **condition** is True. An **ELSE** branch runs when the condition is False. Placing one IF inside another is a **nested** IF. Statements running one after another is **sequence** .

C True or false?

Tick one box · 4 marks

	True	False
In an else if chain, every true branch is executed. *(Only the first true branch runs.)*	☐	☑
else must have a condition. *(else has no condition.)*	☐	☑
Selection uses conditions that evaluate to True or False.	☑	☐

Boundary values should be used to test selection. ☑ ☐

D Exam-style questions

12 marks

1 Describe the difference between sequence and selection. [2]

> Sequence: statements run one after another in order (1) / selection: the program chooses which statements to run depending on whether a condition is True (1)

2 What is output when x = 7? IF x > 10 THEN OUTPUT 'A' ELSE IF x > 5 THEN OUTPUT 'B' ELSE OUTPUT 'C' END IF [1]

> B (1)

3 Write pseudocode that inputs a number of tickets and outputs the total: fewer than 5 tickets cost £8 each, 5 or more cost £6 each. [4]

```
INPUT tickets
IF tickets < 5 THEN
    price ← 8
ELSE
    price ← 6
END IF
total ← tickets * price
OUTPUT total
```

> Marks: input number of tickets (1) / IF tickets < 5 gives £8 (1) / ELSE (or IF tickets >= 5) gives £6 (1) / total calculated with variables and output (1)

4 Explain why the order of conditions matters in an ELSE IF chain. [2]

> Conditions are checked in order and only the first True branch runs; the rest are skipped (1) / so if a broader condition comes first, later branches may never be reached and the wrong result is given (e.g. testing hours <= 5 before hours <= 2) (1)

5 Give three test values for a program where 1–2 hours costs £40/hour, 3–5 hours £35/hour, over 5 hours £30/hour. Explain your choices. [3]

> Any three values, each with justification, e.g.: 2 – upper boundary of the £40 band, expect £80 (1) / 3 – lower boundary of the £35 band, expect £105 (1) / 6 – first value in the £30 band, expect £180 (1). Accept 5 (upper boundary of £35 band, £175) or typical values from each band (e.g. 1, 4, 8) with explanation that each band/branch is tested. One mark per value with a valid reason.

E Quick quiz

Circle one answer · 5 marks

1 x = 15. IF x > 10 AND x < 20 THEN OUTPUT 'Yes' ELSE 'No' [1]

A Yes ✓ B No
C Error D Nothing

2 In C#, which keywords add another condition after an if? [1]

A else **B else if ✓**
C then D elif

3 int score = 50; if (score > 50) "Merit" else if (score >= 40) "Pass" else "Fail" [1]

A Merit **B Pass ✓**
C Fail D Merit Pass

4 An IF inside another IF is... [1]

A a loop **B a nested IF ✓**
C a function D an array

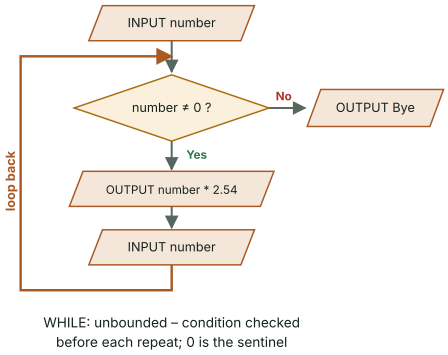
5 Which construct runs statements in order? [1]

A Sequence ✓
B Selection
C Iteration
D Recursion

Iteration: FOR, WHILE and REPEAT loops

D16 ANSWERS

I can: Describe bounded (count-controlled) and unbounded (condition-controlled) iteration · Write FOR and WHILE loops including running totals and counters · Use sentinel values and avoid infinite loops



WHILE: unbounded – condition checked before each repeat; 0 is the sentinel

```
for (int i = 0; i < 5; i++)
```

0 1 2 3 4

FOR: bounded – repeats a set number of times. i < 5 gives 0 to 4, not 5.

do ... while (C#) and REPEAT...UNTIL check at the end: always run once.

Use this diagram to help you: Flowchart of a WHILE loop with a sentinel value, and a FOR loop counting

A Key terms

Write the letter of the matching definition · 6 marks

D **Iteration**	**A** A condition decides which statements run (IF statements).
C **Infinite loop**	**B** Statements executed one after another in order.
E **Variable**	**C** A loop whose condition never becomes false, so it never ends.
F **Constant**	**D** Repeating a block of code (loops).
B **Sequence**	**E** A named memory location whose value can change while the program runs.
A **Selection**	**F** A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

WHILE | iteration | infinite | FOR | sentinel

Repeating code is called **iteration** . A **FOR** loop repeats a set number of times. A **WHILE** loop repeats while a condition is True. A special input value that ends a loop is a **sentinel** . A loop whose condition never becomes False is an **infinite** loop.

C True or false?

Tick one box · 4 marks

	True	False
A WHILE loop always runs at least once. *(It may run zero times; REPEAT always runs once.)*	☐	☑
Running totals should be initialised before the loop.	☑	☐
for (int i = 1; i < 5; i++) runs with i = 5. *(It stops at 4 because the condition is i < 5.)*	☐	☑

REPEAT...UNTIL checks its condition at the end.	☑	☐

D Exam-style questions

13 marks

1 Describe the difference between a FOR loop and a WHILE loop. [2]

> A FOR loop is bounded/count-controlled: it repeats a known/fixed number of times (1) / a WHILE loop is unbounded/condition-controlled: it repeats while a condition is True, checked before each repeat (so it may run zero times) (1)

2 What is output? total ← 0; FOR i ← 1 TO 4: total ← total + i * 2; NEXT i; OUTPUT total [2]

> Working: 2 + 4 + 6 + 8 (1) / output 20 (1)

3 Explain what an infinite loop is and give one cause. [2]

> An infinite loop never ends because its condition never becomes False (1) / cause, e.g. the loop variable is not changed inside the loop, or the condition uses the wrong comparison (1)

4 A loop should end when 0 is entered. It uses WHILE inches > 0. Explain the problem and correct it. [2]

> The condition stops the loop for any value that is not positive, so the loop does not match the specification of ending only when 0 is entered (e.g. a negative value also ends it) (1) / correct to WHILE inches <> 0 (C#: while (inches != 0)) (1)

5 Write pseudocode that inputs numbers until -1 is entered, then outputs how many numbers were entered and their total. [5]

```
count ← 0
total ← 0
INPUT num
WHILE num <> -1
   count ← count + 1
   total ← total + num
   INPUT num
END WHILE
OUTPUT count, total
```

Marks: count and total initialised to 0 (1) / first input before the loop (or REPEAT structure) (1) / loop condition stops at -1 (1) / count and total updated inside the loop, -1 not included (1) / input again inside the loop and both outputs after the loop (1). Accept equivalent C#.

E Quick quiz

Circle one answer · 5 marks

1 Which loop is count-controlled? [1]

A WHILE
B REPEAT UNTIL
C FOR ✓
D IF

2 How many times does for (int i = 3; i < 7; i++) run? [1]

A 3
B 4 ✓
C 5
D 7

3 x ← 10; WHILE x > 0: x ← x - 3; OUTPUT x [1]

A 0
B 1
C -2 ✓
D -1

4 Where should the output of a final total go? [1]

A Inside the loop
B After the loop ✓
C Before the loop
D In the condition

5 A loop that never ends is... [1]

A a nested loop

B an infinite loop ✓

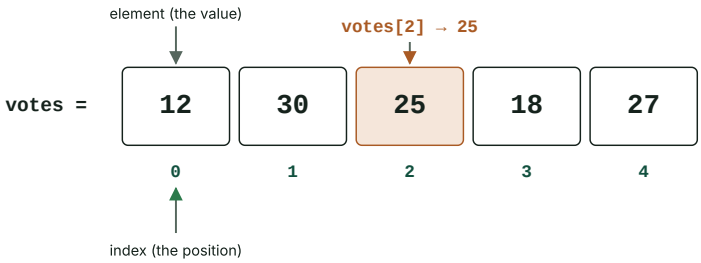
C a bounded loop

D a sentinel

One-dimensional arrays

D17 ANSWERS

I can: Describe an array and its index · Initialise, access, update and loop through an array · Use arrays with running totals, counts, maximum and minimum



Indexes start at 0 · the last index is votes.Length − 1 · loop while i < votes.Length

Use this diagram to help you: An array of five votes showing elements and zero-based indexes

A Key terms

Write the letter of the matching definition · 6 marks

A **Array**	A A data structure holding values of one data type under one name, accessed by index.
C **Index**	B A condition decides which statements run (IF statements).
E **Variable**	C The position number of an element in an array (starting at 0).
F **Constant**	D Statements executed one after another in order.
D **Sequence**	E A named memory location whose value can change while the program runs.
B **Selection**	F A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

element · index · loop · array · zero

An **array** stores many values of the same data type under one name. Each value is an **element** and its position is its **index**. Arrays are **zero**-indexed, so the first element is at position 0. A **loop** is used to process every element.

C True or false?

Tick one box · 4 marks

	True	False
The contents of an array cannot be changed. *(Elements can be updated.)*	☐	☑
The index of an element stores its value. *(The index is its position.)*	☐	☑

In a 6-element array, the last index is 5.	☑	☐
All elements in an array normally have the same data type.	☑	☐

D Exam-style questions

17 marks

1 Describe what is meant by an array. [2]

> A data structure that stores many values of the same data type under one name/identifier (1) / each value (element) is accessed by its index, starting at 0 (1)

2 An array votes holds [14, 22, 9, 31]. What is the value of votes[2]? [1]

> 9 (1)

3 Write a statement that changes the fifth element of the array heights to 162. [2]

> heights[4] ← 162 (C#: heights[4] = 162;). Award: index 4 (zero-indexed fifth element) (1) / correct assignment of 162 to heights (1)

4 Write pseudocode using a loop to set all 30 elements of the array scores to 0. [3]

> FOR i ← 0 TO 29
> scores[i] ← 0
> NEXT i
> Marks: loop structure (1) / correct range 0 to 29 (accept 1 to 30 if consistently 1-indexed) (1) / scores[i] ← 0 using the loop variable (1)

5 Write pseudocode to output the total and average of the 7 values in the array cars. [5]

> total ← 0
> FOR i ← 0 TO 6
> total ← total + cars[i]
> NEXT i
> average ← total / 7
> OUTPUT total, average
> Marks: total initialised to 0 (1) / loop over indexes 0 to 6 (1) / total ← total + cars[i] inside the loop (1) / average ← total / 7 after the loop (1) / total and average output after the loop (1)

6 Complete an algorithm to find the lowest value in the array votes (8 elements). [4]

> lowest ← votes[0]
> FOR i ← 1 TO 7
> IF votes[i] < lowest THEN
> lowest ← votes[i]
> END IF
> NEXT i
> OUTPUT lowest
> Marks: lowest initialised to votes[0] (1) / loop over the remaining elements (1) / compare votes[i] < lowest (1) / replace lowest and output after the loop (1)

E Quick quiz

Circle one answer · 5 marks

1 string[] names = { "Ali", "Bea", "Cal" }; names[1] is... [1]

A Ali — **B Bea ✓**

C Cal — D Error

2 The index of the first element is... [1]

A 1 — **B 0 ✓**

C -1 — D Its value

3 Which loop processes every element of a 10-element array? [1]

A for (int i = 0; i < 10; i++) ✓

B for (int i = 1; i < 10; i++)

C for (int i = 0; i <= 10; i++)

D while (i > 10)

4 To find the largest value you should start with... [1]

A largest = 0 always

B largest = the first element ✓

C largest = the last index

D largest = array.Length

5 int[] arr = { 3, 6, 9 }; arr[0] + arr[2] = [1]

A 9

B 12 ✓

C 15

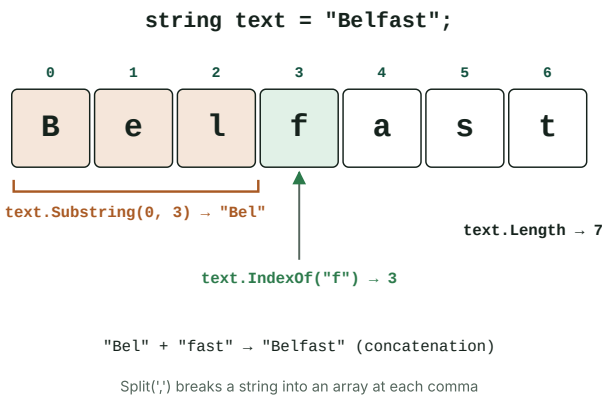
D 18

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

String manipulation

D18 ANSWERS

I can: Use concatenation, Length, indexing and Substring · Split strings and search for characters and substrings · Apply string functions to validation problems



Use this diagram to help you: String indexes, Substring, IndexOf and Length for the word Belfast

A Key terms

Write the letter of the matching definition · 6 marks

C **Concatenation**	**A** Statements executed one after another in order.
D **String function**	**B** A named memory location whose value can change while the program runs.
B **Variable**	**C** Joining strings together.
F **Constant**	**D** A function that manipulates text, e.g. length, split, find, substring.
A **Sequence**	**E** A condition decides which statements run (IF statements).
E **Selection**	**F** A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

concatenation | Length | substring | 0 | Split

Joining two strings is called **concatenation**. The **Length** property gives the number of characters. Part of a string is a **substring**. The **Split** method breaks a string into parts at a separator. The first character of a string is at index **0**.

C True or false?

Tick one box · 4 marks

	True	False
'Hello'[1] returns 'H'. *(It returns 'e'.)*	☐	☑
IndexOf() returns -1 if the substring is not found.	☑	☐
In C#, a number joined to a string with + is converted to text automatically. *("Year " + 11 gives "Year 11".)*	☑	☐

Split(',') removes all the commas and returns one string. *(It returns an array of strings.)* ☐ ☑

D Exam-style questions

10 marks

1 What is meant by concatenation? [1]

> Joining two or more strings together into one (with + in C#) (1)

2 Name a function that returns the number of characters in a string and give an example. [2]

> Length (C#: text.Length) (1) / e.g. "Enniskillen".Length returns 11 (1)

3 word ← 'COMPUTER'. What is the substring from index 3 to index 5 inclusive? [1]

> "PUT" (C O M P U T E R indexed 0–7: index 3 = P, 4 = U, 5 = T) (1)

4 Explain how Split(',') could be used to process the line "Lisburn,Cat,4". [2]

> Split(',') breaks the string at each comma into an array of parts (1) / parts[0] = "Lisburn", parts[1] = "Cat", parts[2] = "4", so each field can be used separately (e.g. Convert.ToInt32(parts[2])) (1)

5 Describe two string functions that could check a password is at least 8 characters and contains @. [4]

> Length (1) – returns the number of characters, so password.Length >= 8 checks it is at least 8 characters (1) / Contains("@") (1) – returns True if '@' appears in the password (accept IndexOf("@") != -1) (1)

E Quick quiz

Circle one answer · 5 marks

1 "Bel" + "fast" = [1]

A Bel fast — **B Belfast ✓**
C fastBel — D Error

2 "Derry".Length = [1]

A 4 — **B 5 ✓**
C 6 — D 0

3 "Antrim".Substring(2, 2) = [1]

A nt — **B tr ✓**
C tri — D ntr

4 "a,b,c".Split(',') gives... [1]

A "abc" — **B an array {"a", "b", "c"} ✓**
C 3 — D {"a,b,c"}

5 "jo@ni.uk".Contains("@") returns... [1]

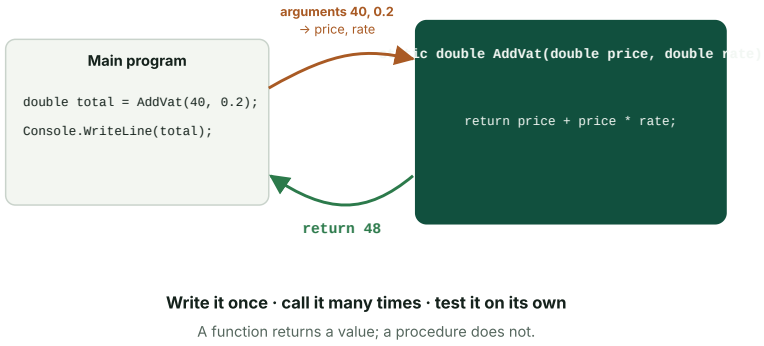
A true ✓ — B false
C 2 — D -1

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Subroutines: user-defined functions and procedures

D19 ANSWERS

I can: Define subroutine, procedure and function · Write functions with parameters and return values · Explain the benefits of user-defined functions



Write it once · call it many times · test it on its own

A function returns a value; a procedure does not.

Use this diagram to help you: A main program passing arguments into a function and receiving the return value

A Key terms

Write the letter of the matching definition · 6 marks

C **Function**

A **Procedure**

B **Parameter**

F **Variable**

D **Constant**

E **Sequence**

A A named subroutine that performs a task without returning a value.

B A value passed into a subroutine.

C A named subroutine that performs a task and returns a value.

D A named value that stays the same while the program runs.

E Statements executed one after another in order.

F A named memory location whose value can change while the program runs.

B Fill the gaps

Use the word bank · 5 marks

procedure | subroutine | reusable | function | parameters

A named block of code that performs one task is a **subroutine** . A **function** returns a value but a **procedure** does not. Values are passed in using **parameters** . Writing code once and using it many times makes it **reusable** .

C True or false?

Tick one box · 4 marks

	True	False
A procedure always returns a value. *(A function returns a value.)*	☐	☑
Functions can be tested on their own.	☑	☐
Console.WriteLine() is a user-defined function. *(It is built in.)*	☐	☑

Parameters allow data to be passed into a function. ☑ ☐

D Exam-style questions

12 marks

1 Define the term function. [2]

> A named, self-contained block of code that carries out a specific task (1) / and returns a single value to the code that called it (1)

2 Describe the difference between a function and a procedure. [2]

> A function returns a value to the calling code (1) / a procedure carries out a task but does not return a value (void in C#) (1)

3 What is a parameter? [1]

> A value/variable passed into a subroutine that it needs to carry out its task, listed in its definition (1)

4 Give two benefits of using user-defined functions. [4]

> Two benefits, each stated (1) + expanded (1), e.g.: code is written once and reused (1) – shorter programs with less repetition (1) / easier to test and debug (1) – each function can be tested on its own (1) / easier to maintain (1) – a change is made in one place only (1) / supports decomposition (1) – different programmers can write different functions (1) / easier to read (1) – the function name describes the task (1)

5 Write a C# function called Area that takes width and height as parameters and returns the area. [3]

> ```
> static double Area(double width, double height)
> {
>     return width * height;
> }
> ```
> Marks: header named Area with two typed parameters width and height (1) / return type double (accept int) (1) / return width * height (1)

E Quick quiz

Circle one answer · 5 marks

1 A subroutine that returns a value is a... [1]

A procedure — **B function ✓**
C loop — D constant

2 In static int Total(int a, int b), a and b are... [1]

A return values — **B parameters ✓**
C constants — D indexes

3 Which is a benefit of functions? [1]

A Code is repeated many times — **B Each part can be tested separately ✓**
C Programs run without a CPU — D They remove the need for variables

4 Which keyword sends a value back from a C# function? [1]

A Console.WriteLine — **B return ✓**
C void — D static

5 Math.Round() is an example of a... [1]

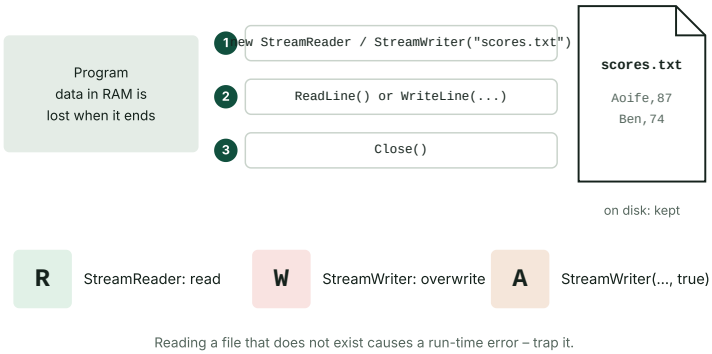
A user-defined function — **B built-in function ✓**
C procedure — D class

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Basic file handling

D20 ANSWERS

I can: Explain why programs store data in files · Open, write to, append to, read and close a text file · Process records read from a file



Use this diagram to help you: Opening, reading or writing, and closing a text file, with read, write and append modes

A Key terms

Write the letter of the matching definition · 6 marks

D **File handling**	**A** Repeating a block of code (loops).
B **Variable**	**B** A named memory location whose value can change while the program runs.
E **Constant**	**C** A condition decides which statements run (IF statements).
F **Sequence**	**D** Opening, reading, writing, appending and closing files to store data permanently.
C **Selection**	**E** A named value that stays the same while the program runs.
A **Iteration**	**F** Statements executed one after another in order.

B Fill the gaps

Use the word bank · 5 marks

closed | persistent | append | opened | Write

Files provide **persistent** storage so data is not lost when a program ends. A file must be **opened** before use and **closed** afterwards. **Write** mode overwrites existing data but **append** mode adds to the end.

C True or false?

Tick one box · 4 marks

	True	False
Opening a file in write mode keeps the existing contents. *(It overwrites them.)*	☐	☑
Files should be closed after use.	☑	☐
Reading a missing file causes a run-time error.	☑	☐

Data in RAM is kept after the program ends. *(RAM is volatile.)*	☐	☑

D Exam-style questions

13 marks

1 Explain why a program might store data in a file. [2]

> Data in variables is held in RAM, which is volatile, so it is lost when the program ends (1) / a file stores data permanently (persistent, non-volatile storage) so it can be loaded next time or shared with other programs/users (1)

2 Name and describe two file handling functions other than read and write. [4]

> Two functions, each named (1) + described (1), e.g.: open (1) – opens or creates a file ready for reading/writing in a chosen mode (1) / close (1) – closes the file so data is saved and the file is not left locked (1) / append (1) – adds data to the end of a file without deleting existing data (1) / delete (File.Delete) (1) – removes the file (1) / exists (File.Exists) (1) – checks whether a file exists before opening (1)

3 Describe the difference between write mode and append mode. [2]

> Write mode creates a new file or overwrites (deletes) all existing data in the file (1) / append mode adds new data to the end of the existing file, keeping existing data (1)

4 Write the three steps needed to add a new score to an existing file. [3]

> Open the file in append mode (C#: new StreamWriter("scores.txt", true)) (1) / write the new score to the file (WriteLine) (1) / close the file (1)

5 What type of error occurs if a program tries to read a file that does not exist? How can it be handled? [2]

> An execution (run-time) error – FileNotFoundException – and the program crashes (1) / handle it with error trapping (try ... catch) or check File.Exists() before opening (1)

E Quick quiz

Circle one answer · 5 marks

1 Which C# statement opens scores.txt to add data to the end? [1]

A new StreamReader("scores.txt") — B new StreamWriter("scores.txt")
C new StreamWriter("scores.txt", true) ✓ — D File.Delete("scores.txt")

2 Why are files used? [1]

A To make programs faster — **B To store data permanently ✓**
C To translate code — D To create loops

3 What should happen after writing to a file? [1]

A Delete it — **B Close it ✓**
C Compile it — D Reopen it in w mode

4 Opening an existing file with new StreamWriter("data.txt") will... [1]

A add to it — **B overwrite it ✓**
C read it — D lock it forever

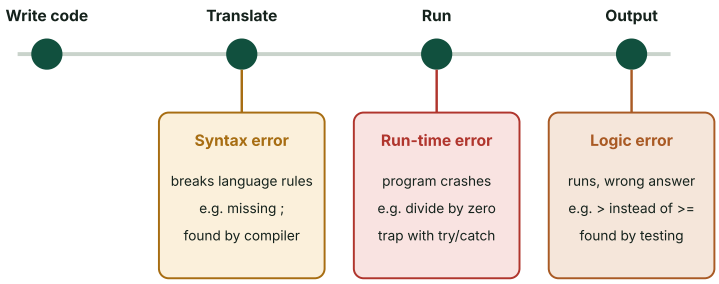
5 Which splits a CSV line into fields? [1]

A line.Split(',') ✓ — B line.Length
C sr.Close() — D line.ToUpper()

Validation, error types and error trapping

D21 ANSWERS

I can: Describe presence, length, type, format and range checks and write validation loops · Distinguish syntax, execution (run-time) and logic errors and how each is found · Use try ... catch to trap errors



Dry runs, trace tables and breakpoints help find logic errors.

Use this diagram to help you: When syntax, run-time and logic errors appear during development

A Key terms

Write the letter of the matching definition · 4 marks

A Syntax error

B Execution / run-time error

D Logic error

C Error trapping

A Breaking the grammar rules of the language; found by the translator before running.

B An error that occurs while the program runs and causes it to crash, e.g. divide by zero.

C Code that detects a run-time error and handles it instead of crashing, e.g. try/except.

D The program runs but gives incorrect results because the reasoning is wrong.

B Fill the gaps

Use the word bank · 6 marks

syntax | Error trapping | logic | range | presence | execution

A **presence** check makes sure a field is not empty. A **range** check makes sure a value is between limits. A **syntax** error breaks the rules of the language. A program that crashes while running has an **execution** error. A program that runs but gives the wrong answer has a **logic** error. **Error trapping** stops a program crashing.

C True or false?

Tick one box · 4 marks

	True	False
Validation can tell if data is correct. *(Only that it is sensible.)*	☐	☑
Dividing by zero causes a run-time error.	☑	☐
A logic error stops the program from running. *(It runs but gives wrong output.)*	☐	☑

A FOR loop is the best choice for a validation loop. *(Use WHILE – the number of attempts is unknown.)* ☐ ☑

D Exam-style questions

22 marks

1 Explain why validation cannot guarantee that data is correct. [2]

> Validation only checks that data is sensible/follows set rules (e.g. within a range) (1) / data can be valid but still wrong, e.g. an age of 21 typed instead of 22 passes a range check (1)

2 Describe a format check and give an example in a password system. [2]

> A format check ensures the data matches a set pattern (1) / e.g. the password must end with a symbol, or contain '@' (1)

3 Write pseudocode that repeatedly asks for 'Y' or 'N' until one of them is entered, showing an error message for invalid input. [5]

> INPUT answer
> WHILE answer <> "Y" AND answer <> "N"
> OUTPUT "Error: please enter Y or N"
> INPUT answer
> END WHILE
> Marks: initial input (1) / condition-controlled loop (WHILE or REPEAT...UNTIL) (1) / correct condition using AND (or UNTIL answer = "Y" OR answer = "N") (1) / error message shown only for invalid input, inside the loop (1) / input again inside the loop (1). Accept equivalent C#.

4 Give one example each of a syntax error, an execution error and a logic error. [3]

> Syntax error, e.g. a missing semicolon or misspelt keyword (Consle) (1) / execution error, e.g. dividing by zero, opening a missing file, index out of range (1) / logic error, e.g. using > instead of >=, or a wrong formula (1)

5 Describe error trapping and one way it improves a program. [4]

> Error trapping (exception handling) catches run-time errors as they happen (1) / using try ... catch, handling the exception instead of crashing (1) / it improves the program because it does not crash – more robust (1) / and the user is given a clear message explaining how to fix the input, so data/work is not lost (1)

6 Describe execution errors and logic errors and how each can be dealt with. (QWC) [6]

> Indicative content – Execution (run-time) errors: occur while the program runs and cause it to crash; examples: dividing by zero, opening a file that does not exist, array index out of range, letters entered where a number is expected (FormatException); dealt with by error trapping (try ... catch) with a helpful message, validation of input, checking File.Exists() first; the error message gives the type and line. Logic errors: the program runs but gives wrong output; examples: wrong formula, > instead of >=, int / int losing the decimal, wrong loop condition; dealt with by testing with expected results (valid, invalid, extreme data), dry runs and trace tables, debugging tools such as breakpoints, stepping and watching variables; fix one error at a time and retest. Levels (QWC): Level 3 (5–6 marks): both error types clearly described with examples and appropriate ways to deal with each; uses technical terms accurately; well-organised, fluent answer with few errors in spelling, punctuation and grammar. Level 2 (3–4 marks): both described with limited methods, or one fully explained; some technical terms; reasonably organised with some errors. Level 1 (1–2 marks): basic points about one or both errors; little or no technical vocabulary; poorly organised with frequent errors. 0 marks: nothing creditworthy.

E Quick quiz

Circle one answer · 5 marks

1 A missing semicolon at the end of a C# statement is a... [1]

A logic error — **B syntax error ✓**
C run-time error — D overflow

2 Which error is found by comparing output with expected results? [1]

A Syntax — **B Logic ✓**
C Compile — D Type

3 Checking that a password is at least 8 characters is a... [1]

A presence check
B length check ✓
C range check
D type check

4 Which C# construct traps run-time errors? [1]

A if/else
B try/catch ✓
C foreach
D static/return

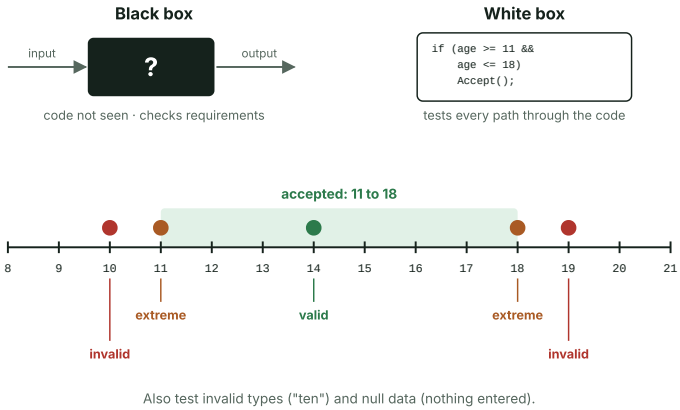
5 while (x < 1 || x > 10) is a validation loop for... [1]

A 1 to 10 inclusive ✓
B below 1 only
C over 10 only
D exactly 10

Testing approaches, test plans and test data

D22 ANSWERS

I can: Describe white box, black box, unit, integration and system testing · Explain iterative testing · Create a test plan with valid, invalid, extreme and null test data



Also test invalid types ("ten") and null data (nothing entered).

Use this diagram to help you: Black box and white box testing, and test data on a number line for an 11 to 18 age range

A Key terms

Write the letter of the matching definition · 6 marks

A White box testing

D Black box testing

B Unit testing

F Integration testing

C System testing

E Test plan

A Testing the internal code, paths and logic, usually by the programmer.

B Testing a single module, function or method on its own.

C Testing the complete, integrated system against its specification.

D Testing inputs and outputs against requirements without seeing the code.

E Table listing each test, its purpose, test data, expected outcome, actual outcome and action.

F Testing that separately built units work together.

B Fill the gaps

Use the word bank · 5 marks

White box | Black box | extreme | unit | integration

White box testing looks at the internal code. **Black box** testing checks inputs and outputs without seeing the code. Testing one module on its own is **unit** testing. Checking modules work together is **integration** testing. Data at the limits of a range is **extreme** data.

C True or false?

Tick one box · 4 marks

	True	False
Black box testing requires programming knowledge. *(It does not.)*	☐	☑
System testing checks the whole system against its requirements.	☑	☐

Null test data means entering nothing.	☑	☐
For a range 1–100, 100 is invalid data. *(100 is extreme (valid) data.)*	☐	☑

D Exam-style questions

17 marks

1 Describe two differences between black box and white box testing. [4]

> Two differences, each stated for both (2 marks each), e.g.: white box testing looks at the internal code, logic and structure (1) – black box testing looks only at inputs and outputs without seeing the code (1) / white box is usually done by the programmer (1) – black box by independent testers or end users (1) / white box aims to test every path, condition and line (e.g. with dry runs/trace tables) (1) – black box checks the functions meet the requirements using a test plan of inputs and expected outputs (1)

2 Describe unit testing. [2]

> Testing one module/function/unit on its own (1) / done first, by the developer, before it is combined with other parts (1)

3 Describe the difference between integration testing and system testing. [2]

> Integration testing checks that separately built units work correctly together (the interfaces between them) (1) / system testing checks the complete system against its specification/requirements, using the test plan (1)

4 A program accepts a number of frames from 10 to 40. Give one example each of valid, extreme and invalid test data. [3]

> Valid: any value 11–39, e.g. 25 (1) / extreme: 10 or 40 (1) / invalid: e.g. 9, 41, -5 or 'ten' (1)

5 Complete a test plan row for testing a blank name entry: purpose, test data, type and expected outcome. [4]

> Purpose: to check that a blank name is rejected (presence check) (1) / test data: nothing entered (blank/null) (1) / type: null (accept invalid/erroneous) (1) / expected outcome: specific error message, e.g. 'Name must be entered', and the user is asked to re-enter (1)

6 Explain what is meant by iterative testing. [2]

> Testing is carried out throughout development, not just at the end (1) / each part is coded, tested, fixed and retested before moving on (1)

E Quick quiz

Circle one answer · 5 marks

1 Testing every path through the code is... [1]

A black box — **B white box ✓**

C beta — D alpha

2 For an age range 11–18, 18 is... [1]

A valid normal — **B extreme ✓**

C invalid — D null

3 Testing modules work together is... [1]

A unit — **B integration ✓**

C system — D acceptance

4 Which column records what actually happened? [1]

A Expected outcome — **B Actual outcome ✓**

C Test data — D Purpose

5 Usual order of testing: [1]

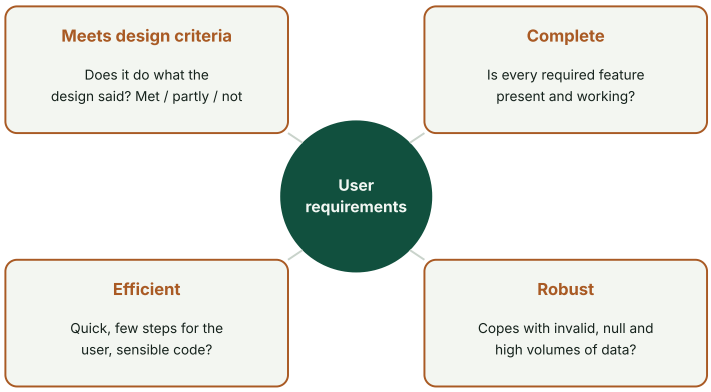
A system → unit → integration — **B unit → integration → system ✓**

C integration → system → unit D system only

Evaluating a solution

D23 ANSWERS

I can: Explain the purpose of evaluation during and after development · Evaluate a solution against design criteria, completeness, efficiency and robustness · Use user requirements and test evidence in an evaluation



Use this diagram to help you: Evaluating a solution against the user requirements: meets design criteria, complete, efficient, robust

A Key terms

Write the letter of the matching definition · 4 marks

A Evaluation

B User requirements

D Robustness

C Stakeholders

A Judging the solution against the user requirements and design.

B The agreed list of what the system must do, used to judge the solution.

C People with an interest in the system: client, end users, developers.

D The ability to handle invalid, unexpected and high volumes of data without failing.

B Fill the gaps

Use the word bank · 5 marks

robust | user requirements | throughout | complete | efficient

Evaluation checks a solution against the original **user requirements**. A **complete** solution has all required features working. A **robust** solution copes with invalid data without crashing. An **efficient** solution does not waste time or resources. Evaluation should take place **throughout** development.

C True or false?

Tick one box · 4 marks

	True	False
Evaluation only happens after the program is finished. *(Also during development.)*	☐	☑
Robustness testing should use high volumes of data.	☑	☐
A list of features is the same as an evaluation. *(Evaluation judges how well with evidence.)*	☐	☑

Stakeholders include the end users of the system. ☑ ☐

D Exam-style questions

10 marks

1 Give two reasons why a solution should be evaluated. [2]

> Any two from: to judge how well the solution solves the original problem (1) / to check it meets the user requirements/design criteria (1) / to identify weaknesses, missing features and improvements/refinements (1) / to check it is complete, efficient and robust (1)

2 Explain how user requirements are used during evaluation. [2]

> The user requirements are the agreed standard the solution is measured against – each is checked as a checklist: fully met / partly met / not met (1) / if a feature is missing, the requirements show what must be added before the solution is complete (1)

3 What is meant by a robust solution? [2]

> A robust solution handles errors and unexpected input without crashing (1) / it copes with invalid, unexpected/exceptional and high volumes of data (e.g. through validation and error trapping) (1)

4 Describe how a developer could test the robustness of a booking system. [2]

> Test it with invalid, extreme/exceptional and null data (e.g. a booking of 0 or -2 nights, a blank name) (1) / and with high volumes of data (many bookings at once), checking it does not crash and gives suitable error messages (1). Must refer to at least two kinds of data.

5 Explain what is meant by an efficient solution. [2]

> It works quickly without wasting resources (processing time, memory, storage) (1) / and with little effort from the user (few steps), using sensible code such as loops and functions (1)

E Quick quiz

Circle one answer · 5 marks

1 Evaluation measures a solution against... [1]

A the CPU speed
B the user requirements ✓
C the number of lines of code
D the font used

2 A solution that copes with invalid input without crashing is... [1]

A efficient
B robust ✓
C complete
D abstract

3 Testing with very large amounts of data checks... [1]

A syntax
B robustness ✓
C abstraction
D compilation

4 A complete solution... [1]

A has every required feature working ✓
B has no comments
C uses only FOR loops
D is written in Java

5 Which is an evaluative comment? [1]

A The program has a menu.
B Requirement 2 is fully met: all 10 test bookings were stored correctly. ✓
C I used C#.
D The program is in a file.